

Chapter Three

INTRODUCTION TO SOFTWARE ENGINEERING

Software Engineering

- ***Software Engineering*** is composed of two words, **software** and **engineering**.
 - **Software** is more than just a program code
 - ✓ A *program* is an executable code, having some computational purpose
 - ✓ *Software* is a collection of executable programming code, associated libraries and documentations
 - ✓ Software, when made for a specific requirement is called **software product**.

- **Engineering** is developing products, using well-defined, scientific principles and methods.
- ***Software Engineering*** is an engineering branch associated with the development of software product using well-defined scientific principles, methods and procedures with the outcome of an efficient and reliable software product.

Software engineering principles use two important techniques to reduce problem complexity:

❖ *Abstraction*: simplifying problems by omitting irrelevant details.

❖ *Decomposition*: dividing complex problems into several smaller problems and solving smaller problems one by one

Software Engineering as an Engineering discipline

- Apply theories, methods, and tools and use them selectively
- Try to discover solutions to problems even when there are no applicable theories and methods.
- Find solutions within constraints
- Software engineering is concerned about software project management and the development of tools, methods, and theories in addition to technical process of software development.

SE Vs CS

- **CS** is concerned with the theories and methods that underlie **computers and software systems** whereas SE is concerned with the practical problems of producing software.
- Some knowledge of CS is essential for SE like some knowledge of physics is essential for electrical engineers.

SE Vs System Engineering

- System engineering is concerned with all aspects of the development and evolution of systems
 - Hardware development,
 - Policy and process design and
 - System deployment, as well as software engineering
- System Engineers specify the **system**, define its overall **architecture**, and **integrate** the different parts to create the finished system.
- But SE mainly concerned on the development of **software products**.

PRESENTATION TIME